

Hasan Bukhari

Computational Biology • Bioinformatics • Research Engineering

+1 601 913 4692 | hasan.bukhari25@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Southern Mississippi

B.S. in Computer and Information Sciences; Minor in Biological Sciences

GPA: 4.00 • President's List • Academic Excellence Scholar

Expected Dec 2028

Hattiesburg, MS

Beaconhouse College Program

*Cambridge A Levels: 4 A*s in Biology, Chemistry, Mathematics, and Physics*

May 2024

Karachi, Pakistan

RESEARCH & EXPERIENCE

Data Analyst Intern, Health, Safety & Environment

Engro Technical Services

Jul 2026 – Aug 2026

Karachi, Pakistan

- Built executive Power BI dashboards and automated reporting workflows across 130,000+ incident, near-miss, inspection, audit, and corrective-action records.
- Developed Python, Power Query, and FastAPI pipelines for overdue-action risk scoring and scheduled reporting; automated daily change monitoring across approximately 1,750 engineering and HSE files.

Independent Computational Biology Researcher

CancerShift and Chromatin Contact Prediction Across Chromosomes

2026 – Present

Independent

- **CancerShift:** Developed a leakage-controlled multicancer classification pipeline spanning TCGA, GEO, and CPTAC-GDC cohorts; achieved 99.59% accuracy on a blind microarray cohort and 98.24% on external RNA-seq data.
- **Chromatin contact prediction:** Designed and trained A17, a seven-track residual neural network for predicting Hi-C contact maps from epigenomic signals under strict chromosome-level holdouts, frozen decisions, and a locked test set.
- Found that relative chromatin structure generalized across unseen chromosomes more consistently than absolute contact-map scale; documented negative results and failed target-blind calibration in a manuscript in preparation.

Undergraduate Researcher, Ichthyology Lab

University of Southern Mississippi

Spring 2026 – Present

Hattiesburg, MS

- Analyze chromosome-scale *Fundulus* genome assemblies across two related species and 24 pseudochromosome labels to evaluate synteny, correspondence, and candidate fusion/fission patterns.
- Generate alignment-based correspondence matrices and technical reports translating genomic methods, limitations, and next-step recommendations for lab review.

Student Worker, Operations Technology & Event Support

Thad Cochran Center / Union Operations, University of Southern Mississippi

Jun 2024 – Present

Hattiesburg, MS

- Coordinate technical support and event operations; developed SetItUp and UnionInventoryPro for room requests, layouts, equipment tracking, and inventory audits.

Software Engineering Intern

Shahid I.T. Consultants

Aug 2023 – Feb 2024

Karachi, Pakistan

- Automated recurring record-update workflows in Python, reducing manual workload by 30%; supported integration testing and deployment documentation.

SELECTED RESEARCH SOFTWARE

BioLitGraph | Python, Flask, PubTator, NetworkX | [GitHub](#)

2026

- Built a biomedical literature-mining application with entity extraction, interactive relationship graphs, and exportable evidence tables.

QTL Analysis Toolkit | Python, pandas, Matplotlib | [GitHub](#)

2025

- Implemented LOD scans, permutation thresholds, significant-marker exports, and publication-ready statistical-genomics plots.

TECHNICAL SKILLS

Programming & Modeling: Python, SQL, C++, R; PyTorch, scikit-learn, pandas, NumPy, SciPy, Biopython

Computational Biology: RNA-seq, microarrays, Hi-C, ATAC-seq, epigenomics, statistical genomics, synteny, QTL analysis

Research Engineering & Analytics: Linux/WSL, Git/GitHub, FastAPI, Flask, Power BI, DAX, Power Query, Excel

Data Resources: TCGA, GEO, CPTAC, ENCODE, Ensembl, PubMed/PubTator

LEADERSHIP & HONORS

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and club operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.